# Adversarial Autoencoders

From VAEs and GANs to adversarial inference

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# Outline

1. VAE
2. GAN
3. $f$-divergences
4. Forward vs reverse KL
5. Adversarial autoencoders
6. Adversarial Variational Bayes
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# VAE

# Log-likelihood and the ELBO

$$L(\theta) = \sum_{i=1}^{N} \log p(x^{(i)}|\theta) = \mathcal{L}(q(Z), \theta) + KL\big(q(Z) \,\|\, p(Z|X, \theta)\big) \;\geq\; \mathcal{L}(q(Z), \theta),$$

where

$$\mathcal{L}(q(Z), \theta) = \sum_{i=1}^{N} \mathcal{L}(q^{(i)}(z), \theta) = \sum_{i=1}^{N} \int q^{(i)}(z) \log \frac{p(x^{(i)}, z|\theta)}{q^{(i)}(z)} \mathrm{d}z$$

and

$$KL\big(q(Z) \,\|\, p(Z|X, \theta)\big) = \sum_{i=1}^{N} KL\big(q^{(i)}(z) \,\|\, p(z|x^{(i)}, \theta)\big) = \sum_{i=1}^{N} \int q^{(i)}(z) \log \frac{q^{(i)}(z)}{p(z|x^{(i)}, \theta)} \mathrm{d}z.$$

# Amortized inference

In the case of amortized inference,

$$q^{(i)}(z) \approx q(z|x^{(i)}, \varphi) = \prod_{j=1}^{d} N\big(z_j \mid m_j(x^{(i)}),\ s_j^2(x^{(i)})\big),$$

and the ELBO becomes

$$\mathcal{L}(\theta, \varphi) = \sum_{i=1}^{N} \Big[\mathbb{E}_{q(z|x^{(i)},\varphi)} \log p(x^{(i)}|z, \theta) - KL\big(q(z|x^{(i)}, \varphi) \,\|\, p(z)\big)\Big] \to \max_{\theta,\varphi}.$$

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**Which divergence does MLE minimize?**
As $N \to \infty$,

$$\frac{1}{N}\sum_i \log p(x^{(i)}|\theta) \ \to\ \mathbb{E}_{p_{\text{data}}} \log p(x|\theta) = -KL(p_{\text{data}} \| p_\theta) - H(p_{\text{data}}),$$

so MLE (and the VAE, through its lower bound) minimizes the *forward* divergence $KL(p_{\text{data}} \| p_\theta)$.

# GAN

# Representation theorem

**Theorem (Kolmogorov, von Neumann)**
For a Borel probability measure $\mu$ on a complete separable metric space $X$ there exists a random variable $\xi\colon [0,1] \to X$ on $([0,1],\ \mathcal{B}([0,1]),\ \lambda)$ with distribution $P_\xi = \mu$.

For $X = \mathbb{R}$ this is just the quantile transform: $\xi(u) = F^{-1}(u)$, where $F$ is the c.d.f. of $\mu$.

**Idea:** approximate such a map by a neural network, $x = G(\xi)$, $\xi \sim r$, and call it the generator. Denote the resulting distribution $G_\sharp r$ by $q(x)$.

**Remark.** One uniform scalar suffices in theory, but the map is then wildly discontinuous. A network is continuous, so $G(\mathbb{R}^d)$ is connected: it cannot put mass exactly on disconnected modes. In practice $\xi \sim N(0, I_d)$, $d > 1$.

A. N. Kolmogorov, *Grundbegriffe der Wahrscheinlichkeitsrechnung*, Springer, 1933. J. von Neumann, Zur Operatorenmethode in der klassischen Mechanik, *Ann. Math.* 33(3):587–642, 1932. O. Kallenberg, *Foundations of Modern Probability*, 2nd ed., Springer, 2002, Ch. 3.

# Why not MSE?

MSE is not a very good perceptual metric (see the examples in homework 2).

With a Gaussian decoder $p(x|z,\theta) = N\big(x \,|\, \mathrm{DEC}_\theta(z),\ \sigma^2 I\big)$,

$$\log p(x|z,\theta) = -\frac{1}{2\sigma^2}\,\|x - \mathrm{DEC}_\theta(z)\|_2^2 + \mathrm{const},$$

so the reconstruction term of the ELBO is an expected MSE.

- With a deterministic encoder and no KL term it is the plain autoencoder loss.
- Even with the exact posterior $q(z|x^{(i)},\varphi) = p(z|x^{(i)},\theta)$, the gap vanishes ($\mathcal{L} = \log p(x^{(i)}|\theta)$), but the decoder is still judged pixel by pixel.

## From a reconstruction loss to a discriminator

Instead of

$$\min \ \|\mathrm{DEC}(z) - x\|_2^2 \qquad \text{we look for} \qquad \max \ D\big(\mathrm{DEC}(z)\big),$$

where $D(x)$ is the probability that $x$ belongs to the reference dataset.

- $D =$ encoder network $+$ linear layer $+$ sigmoid, as in binary classification (VAE-GAN, Larsen et al., 2016).

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- $D$ = encoder network + linear layer + sigmoid, as in binary classification (VAE-GAN, Larsen et al., 2016).
- $D$ does *not* compare the output with a specific reference image $x$.
- So the encoder is no longer needed to produce $z$: sample $z = \xi \sim r$, and $\mathrm{DEC}$ becomes the generator $G$.

# GAN objectives

**Discriminator:**

$$\frac{1}{N}\sum_{i=1}^{N}\log D(x^{(i)}) + \frac{1}{M}\sum_{j=1}^{M}\log\left(1 - D(G(\xi^{(j)}))\right) \to \max_{D}$$

**Generator:**

1. minimax (saturating):

$$-\frac{1}{M}\sum_{j=1}^{M}\log\left(1 - D(G_\theta(\xi^{(j)}))\right) \to \max_{\theta}$$

2. non-saturating heuristic (Goodfellow et al., 2014):

$$\frac{1}{M}\sum_{j=1}^{M}\log D(G_\theta(\xi^{(j)})) \to \max_{\theta}$$

# Saturation

- The minimax loss saturates at the beginning of training: when $G$ is bad, $D(G(\xi)) \approx 0$ and $\log(1 - D)$ is flat, so the gradients vanish.
- In practice everyone uses the non-saturating loss.
- One can also maximize the sum of the two, so that the gradients are non-zero both at the initial and at the terminal stages of training:

$$\mathbb{E}_{\xi \sim r}\Big[\log D(G_\theta(\xi)) - \log\big(1 - D(G_\theta(\xi))\big)\Big] \to \max_\theta.$$

# Optimal discriminator

For a fixed generator,

$$D^\star(x) = \frac{p(x)}{p(x) + q(x)}.$$

*Why?* Maximize $p(x) \log D + q(x) \log(1 - D)$ pointwise in $D \in (0, 1)$.

# Optimal discriminator

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*Why?* Maximize $p(x) \log D + q(x) \log(1 - D)$ pointwise in $D \in (0, 1)$.

Substituting $D^\star$ into the (summed) generator functional:

$$\int r(\xi) \log \frac{D^\star(G_\theta(\xi))}{1 - D^\star(G_\theta(\xi))} \mathrm{d}\xi = \int q(x) \log \frac{p(x)}{q(x)} \mathrm{d}x = -KL(q\|p) \to \max_q .$$

This GAN minimizes the *reverse* $KL(q\|p)$; the VAE minimizes the *forward* $KL(p\|q)$.

## Is substituting $D^\star$ legitimate?

It is an idealization: in training $D$ is held fixed while $\theta$ is updated. It is still correct at the level of gradients, because $\int \nabla_\theta q_\theta = 0$:

$$\nabla_\theta \int q_\theta \log \frac{p}{q_{\theta_0}}\Big|_{\theta=\theta_0} = -\nabla_\theta KL(q_\theta \| p)\Big|_{\theta=\theta_0}.$$

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In the same way (Arjovsky & Bottou, 2017), with $D = D^\star$ the two generator losses separately give

$$\begin{aligned} \text{minimax:} \quad & \nabla_\theta \, 2\, JS(p, q_\theta), \\ \text{non-saturating:} \quad & \nabla_\theta \big[ KL(q_\theta \| p) - 2\, JS(p, q_\theta) \big], \end{aligned}$$

and their sum is $\nabla_\theta KL(q_\theta \| p)$.

# $f$-divergences

# $f$-divergence and its variational form

For a convex $f\colon (0,\infty)\to\mathbb{R}$ with $f(1)=0$,

$$D_f(P\|Q)=\int q(x)\, f\!\left(\frac{p(x)}{q(x)}\right)\mathrm{d}x \;\geq\; 0.$$

**Variational representation** (Nguyen, Wainwright & Jordan, 2010). With the convex conjugate $f^*(t)=\sup_u\{tu-f(u)\}$, we have $f(u)=\sup_t\{tu-f^*(t)\}$ and

$$D_f(P\|Q)=\sup_T\Big\{\mathbb{E}_{x\sim P}\,T(x)-\mathbb{E}_{x\sim Q}\,f^*\big(T(x)\big)\Big\},\qquad T^\star(x)=f'\!\left(\frac{p(x)}{q(x)}\right).$$

Restricting $T$ to a neural network $T_\omega$ gives a lower bound.

$f$-**GAN** (Nowozin, Cseke & Tomioka, 2016):

$$\min_\theta\max_\omega\;\mathbb{E}_{x\sim p}\,T_\omega(x)-\mathbb{E}_{\xi\sim r}\,f^*\big(T_\omega(G_\theta(\xi))\big).$$

## Common $f$-divergences

| divergence | $f(u)$ | $f^*(t)$ | $T^\star$ |
|---|---|---|---|
| $KL(P\|Q)$ | $u \log u$ | $e^{t-1}$ | $1 + \log \frac{p}{q}$ |
| $KL(Q\|P)$ | $-\log u$ | $-1 - \log(-t),\ t < 0$ | $-\frac{q}{p}$ |
| $2\,JS(P,Q)$ | $u \log u - (u+1) \log \frac{u+1}{2}$ | $-\log(2 - e^t),\ t < \log 2$ | $\log \frac{2p}{p+q}$ |
| GAN: $2\,JS - \log 4$ | $u \log u - (u+1) \log(u+1)$ | $-\log(1 - e^t),\ t < 0$ | $\log \frac{p}{p+q}$ |
| Pearson $\chi^2$ | $(u-1)^2$ | $t^2/4 + t$ | $2\left(\frac{p}{q} - 1\right)$ |

For the GAN row put $T = \log D$: then $\mathbb{E}_p T - \mathbb{E}_q f^*(T) = \mathbb{E}_p \log D + \mathbb{E}_q \log(1 - D)$, exactly the discriminator functional, and $T^\star = \log D^\star$.

## Limitation of $f$-divergences

If $p$ and $q$ live on disjoint low-dimensional manifolds, then $p/q$ is 0 or $\infty$ almost everywhere.

- Every $f$-divergence then saturates (e.g. $JS = \log 2$).
- It gives no useful gradient for the generator.

This is the motivation for Wasserstein (optimal transport) distances.

# Forward vs reverse KL

# Mass-covering vs mode-seeking

Let $p$ be the data distribution and $q$ the model.

- **Forward** $KL(p\|q) = \mathbb{E}_p \log \frac{p}{q}$ (MLE, VAE). Infinite if $q \approx 0$ where $p > 0$, so $q$ must cover all modes of $p$.
  *Mass-covering*: samples tend to be blurry, with mass between the modes.
- **Reverse** $KL(q\|p) = \mathbb{E}_q \log \frac{q}{p}$ (the GAN above). Infinite if $q > 0$ where $p \approx 0$, but missing modes are not penalized.
  *Mode-seeking*: samples are sharp, but modes can be dropped (mode collapse).

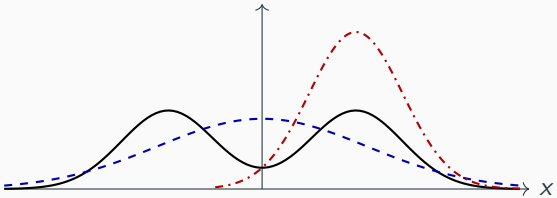
The non-saturating loss behaves like $KL(q\|p) - 2JS$, and the $-2JS$ term does not fix missing modes either: a theoretical explanation of mode dropping in GANs.

# Example: fitting one Gaussian to two modes

$p = \frac{1}{2}N(-\mu, 1) + \frac{1}{2}N(\mu, 1)$,
$q = N(m, s^2)$.

- Forward KL: moment matching, $m = 0$, $s^2 = 1 + \mu^2$. One wide Gaussian between the modes.
- Reverse KL, large $\mu$: $m \approx \pm\mu$, $s^2 \approx 1$. A single mode.

The VAE also uses reverse KL, but only for inference: the encoder minimizes $KL\big(q(z|x,\varphi) \,\|\, p(z|x,\theta)\big)$, so amortized posteriors tend to be under-dispersed.




$p$ (data, $\mu = 2$): solid
forward KL fit $N(0, 5)$: dashed
reverse KL fit $N(2, 1)$: dash-dot

# What adversarial training gives us

The main contributions of adversarial training:

- The discriminator learns the discrepancy between two distributions (a density ratio, or an $f$-divergence), using samples only.
- The generator can be used as a sampler, with no explicit density.

# Adversarial autoencoders

# Motivation: aggregated posterior vs prior

We would like the *aggregated posterior* to match the prior:

$$q(z) = \int p_{\text{data}}(x)\, q(z|x,\varphi)\mathrm{d}x \;\approx\; \frac{1}{N}\sum_{i=1}^{N} q(z|x^{(i)},\varphi) \;\approx\; p(z).$$

**Why is this important?**

- At generation time we sample $z \sim p(z)$, but during training the decoder only sees codes $z \sim q(z)$.
- If $q(z) \neq p(z)$, the prior puts mass on "holes": regions where $q(z)$ is small and the decoder was never trained. There the generator produces unrealistic $x$.
- The ELBO does not enforce $q(z) \approx p(z)$ directly: its KL term acts on each $q(z|x^{(i)},\varphi)$ separately and competes with reconstruction.

**Idea:** train a discriminator to recognize codes that do not come from $p(z)$.

# ELBO surgery

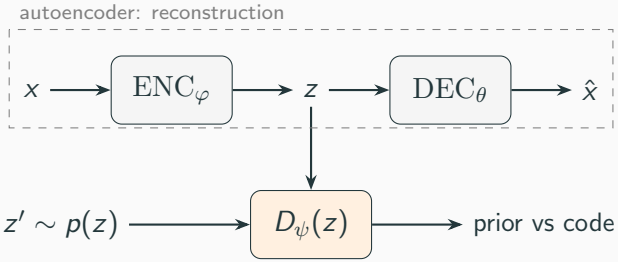
The averaged KL term of the ELBO decomposes as (Hoffman & Johnson, 2016)

$$\frac{1}{N}\sum_{i=1}^{N} KL\big(q(z|x^{(i)},\varphi)\,\|\,p(z)\big) = I_q(i;z) + KL\big(q(z)\,\|\,p(z)\big),$$

where $I_q(i;z) \leq \log N$ is the mutual information between the index $i$ of a data point and its code $z$ under $q$.

- The VAE penalizes both terms.
- Penalizing $I_q$ pushes the codes of different images together: one source of blurriness and of posterior collapse.
- The AAE (Makhzani et al., 2015) keeps only $q(z) \approx p(z)$ and enforces it adversarially, with a GAN in the latent space.

# AAE architecture




New samples: $x = \mathrm{DEC}_\theta(z)$ with $z \sim p(z)$.

# AAE training

Alternate two phases on each minibatch.

1. *Reconstruction:*

$$\min_{\varphi,\theta} \frac{1}{N}\sum_{i=1}^{N} \left\| x^{(i)} - \mathrm{DEC}_\theta\big(\mathrm{ENC}_\varphi(x^{(i)})\big) \right\|_2^2 \quad (\text{or } -\log p(x^{(i)}|z,\theta)).$$

2. *Regularization:* a discriminator $D_\psi(z)$ separates prior samples from codes,

$$\max_{\psi} \mathbb{E}_{z\sim p(z)} \log D_\psi(z) + \frac{1}{N}\sum_{i=1}^{N} \log\Big(1 - D_\psi\big(\mathrm{ENC}_\varphi(x^{(i)})\big)\Big),$$

and the encoder plays the generator:

$$\max_{\varphi} \frac{1}{N}\sum_{i=1}^{N} \log D_\psi\big(\mathrm{ENC}_\varphi(x^{(i)})\big).$$

# Why AAE is useful; variants

## Why it is useful

- Only *samples* from $p(z)$ are needed, never its density: the prior can be a mixture of Gaussians, a swiss roll, or a categorical distribution.
- The encoder may be deterministic: no reparameterization, no analytic KL.
- The encoder may be *implicit*: $z = \mathrm{ENC}_\varphi(x, \varepsilon)$ with noise $\varepsilon$ fed into the network, so $q(z|x, \varphi)$ can be sampled but its density is unknown.

## Variants

- Supervised: the label $y$ is fed to the decoder, so $z$ learns the "style".
- Semi-supervised and clustering: a categorical code $y$ with its own discriminator against $\mathrm{Cat}(K)$, plus a Gaussian style code $z$.
- Dimensionality reduction.

# Adversarial Variational Bayes

# AVB: a discriminator for the density ratio

The AAE drops $I_q$, so it no longer optimizes a bound on $\log p(x|\theta)$. **AVB** (Mescheder, Nowozin & Geiger, 2017) keeps the full ELBO but allows an implicit encoder. The ratio $q/p$ is estimated by a discriminator $D(x, z)$ on *pairs*:

$$\begin{aligned} &\text{class 1:} \quad (x, z) \sim p_{\text{data}}(x)\, p(z), \\ &\text{class 2:} \quad (x, z) \sim p_{\text{data}}(x)\, q(z|x, \varphi), \end{aligned}$$

with $D(x, z)$ the probability of class 1, trained by logistic regression:

$$\max_D \; \mathbb{E}_{p_{\text{data}}(x)} \Big[ \mathbb{E}_{p(z)} \log D(x, z) + \mathbb{E}_{q(z|x,\varphi)} \log \big(1 - D(x, z)\big) \Big].$$

As for the GAN, the optimum is

$$D^\star(x, z) = \frac{p(z)}{p(z) + q(z|x, \varphi)}, \qquad \log \frac{1 - D^\star(x, z)}{D^\star(x, z)} = \log \frac{q(z|x, \varphi)}{p(z)}.$$

# AVB objective

Substituting into the ELBO:

$$\mathcal{L}(\theta, \varphi) = \sum_{i=1}^{N} \left[ \mathbb{E}_{q(z|x^{(i)},\varphi)} \log p(x^{(i)}|z,\theta) - \mathbb{E}_{q(z|x^{(i)},\varphi)} \log \frac{1 - D(x^{(i)}, z)}{D(x^{(i)}, z)} \right] \to \max_{\theta,\varphi}.$$

- Both expectations use reparameterized samples $z = \mathrm{ENC}_{\varphi}(x^{(i)}, \varepsilon)$.
- $D$ can be held fixed when differentiating with respect to $\varphi$, because $\mathbb{E}_q \nabla_{\varphi} \log q(z|x,\varphi) = 0$.
- As in the GAN section, only the pathwise gradient through $z$ remains.

# AVB: problems and adaptive contrast

- *Not a bound.* With $D \neq D^\star$ the objective only estimates the ELBO: it is neither a lower nor an upper bound on $\log p(x|\theta)$.
- *The ratio is hard to estimate.* A good posterior is much narrower than $p(z)$, so $KL(q\|p)$ is large and the classes are almost separable: $D$ saturates and $\log \frac{1-D}{D}$ is poor.

**Adaptive contrast.** Compare $q$ with a tractable Gaussian $r_\alpha(z|x)$ that matches the mean and variance of $q(z|x,\varphi)$:

$$KL\big(q \,\|\, p\big) = \underbrace{KL\big(q(z|x,\varphi) \,\|\, r_\alpha(z|x)\big)}_{\text{adversarial}} + \underbrace{\mathbb{E}_{q(z|x,\varphi)}\big[\log r_\alpha(z|x) - \log p(z)\big]}_{\text{explicit densities}}.$$

Since $r_\alpha \approx q$, the two classes are hard to separate: the regime where the ratio estimate is accurate.

# Summary

# Explicit, latent-variable and implicit models

| | Explicit | Latent-variable (semi-explicit) | Implicit |
|---|---|---|---|
| Example | normalizing flow | VAE | GAN |
| Density | exact $p_\theta(x)$ | $p_\theta(x, z)$ exact; $p_\theta(x)$ only bounded (ELBO, IWAE) | none, samples only |
| Training | exact MLE | ELBO | adversarial |
| Sampling | efficient (one pass)* | efficient (ancestral) | efficient (one pass) |
| Inverse $x \mapsto z$ | exact | approximate (encoder) | none** |
| Latent dimension | $= \dim x$ | any, usually $\ll \dim x$ | any |
| Flexibility | restricted (invertible layers) | less restricted | flexible |

* Coupling flows (RealNVP, Glow); autoregressive flows such as MAF sample one dimension at a time. ** Unless an encoder is trained jointly (BiGAN/ALI).

AAE and AVB sit between the last two columns: the decoder is a latent-variable model, but the encoder, the prior, or both are implicit.

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